INCH-POUND

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4 August 2009
SUPERSEDING
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18 June 1999

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 7402

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: High vacuum clipper diode or rectifier.

Outline: See figure 1.

Mounting: Vertical position, base down.

Weight: 14.3 ounces nominal.

ABSOLUTE RATINGS:

| Parameter:<br>Unit: | Ef<br>V ac | epx<br>kv | ib<br>a | Ib<br>mA dc | Ieff<br>A ac | Pp<br>W |
|---|---|---|---|---|---|---|
| Clipper | | | | | | |
| Maximum: | 12.1 | 33 | 50 | --- | 1.25 | 450 |
| Minimum: | 10.9 | --- | --- | --- | --- | --- |
| Rectifier | | | | | | |
| Maximum: | 12.1 | 33 | 2.7 | 860 | --- | 450 |
| Minimum: | 10.9 | --- | --- | --- | --- | --- |
| Test condition: | 11.5 | --- | --- | --- | --- | --- |

General:

Qualification - Not required.



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TABLE I. Testing and inspection.

| Inspection | Method MIL-STD-1311 | Condition | Acceptance Level | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | | | | Min | Max | |
| Conformance inspection, part 1 | | | | | | | |
| Filament current | 1301 | | 0.65 | If | 14.75 | 16.25 | A ac |
| Peak emission | 1231 | eb = 8.0 kv | 0.65 | is | 30 | --- | a |
| Electrode current (anode) | 1256 | Eb = 330 V dc | 0.65 | Ib | 860 | --- | mA dc |
| Operation of rectifiers | 1353 | epx = 33 kv;<br>ib = 2.7 a<br>Ib = 860 mA dc 1/ | 0.65 | --- | --- | --- | --- |
| Conformance inspection, part 2 | | | | | | | |
| Low-frequency vibration | 1031 | No voltages | 6.5 | --- | --- | --- | --- |
| Shock | 1041 | Hammer angle 25° | --- | --- | --- | --- | --- |
| Operation | --- | epx = 33 kv<br>tp = 1 μsec;<br>Zo = 53 ohms;<br>Rs = 300 ohms (min);<br>Prr = 340<br>R1 = 0 1/ 2/ | --- | ib | 50 | --- | a |
| Conformance inspection, part 3 | | | | | | | |
| Life test provisions | --- | Group D; operation of rectifiers | --- | t | 500 | --- | hr |
| Life-test end points: | --- | | | | | | |
| Peak emission | 1231 | | --- | is | 25 | --- | a |

1/ The tube shall operate for 2 minutes without arcing or signs of gas during a test interval not to exceed 5 minutes.

2/ The tube shall be operated as a clipper diode in a thyratron modulator.



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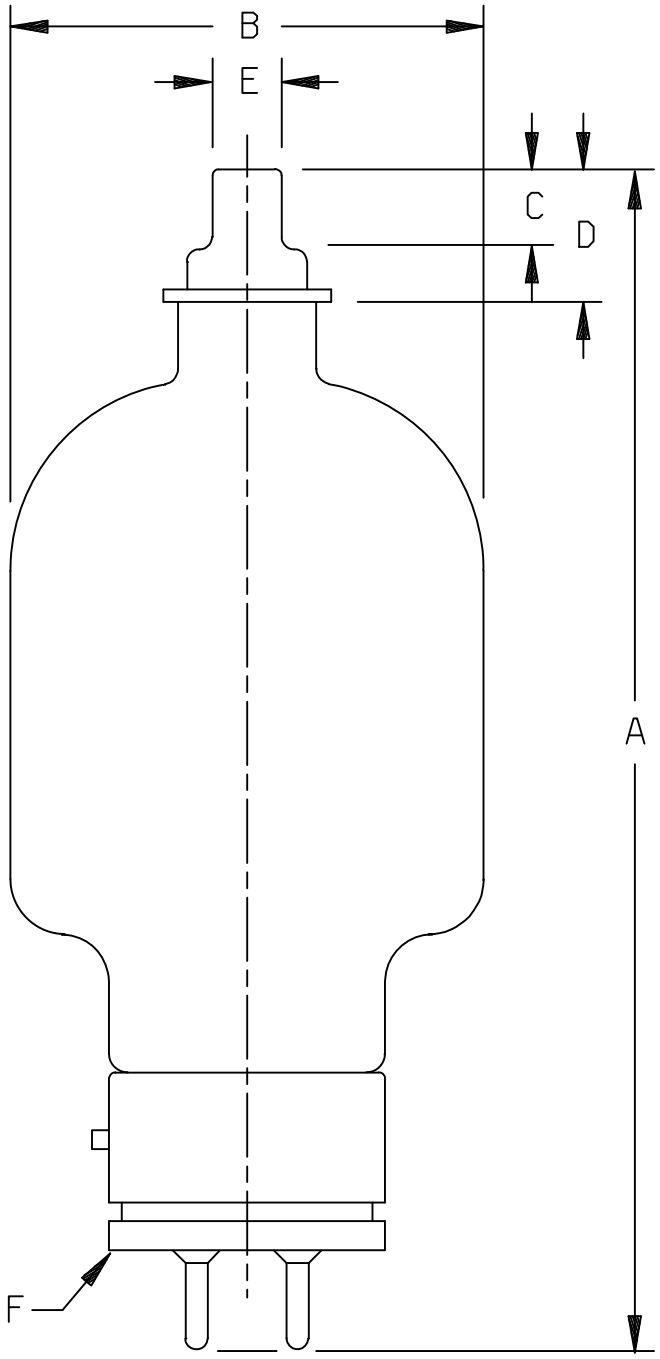


| Dimensions | | | | |
|---|---|---|---|---|
| Ltr | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| Conformance inspection, part 2 | | | | |
| A | 9.250 | 9.750 | 235.15 | 247.65 |
| B | --- | 3.625 | --- | 92.07 |
| C | .594 | .656 | 15.09 | 16.66 |
| Conformance inspection, part 3 (periodic check) | | | | |
| E | .559 | .573 | 14.20 | 14.60 |
| F | Base A-4-18 | | | |
| Reference dimensions | | | | |
| D | 1.00 | | 25.40 | |

FIGURE 1. Outline drawing of tube type 7402.



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Referenced documents. In addition to MIL-PRF-1, this document references the following:
MIL-STD-1311

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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